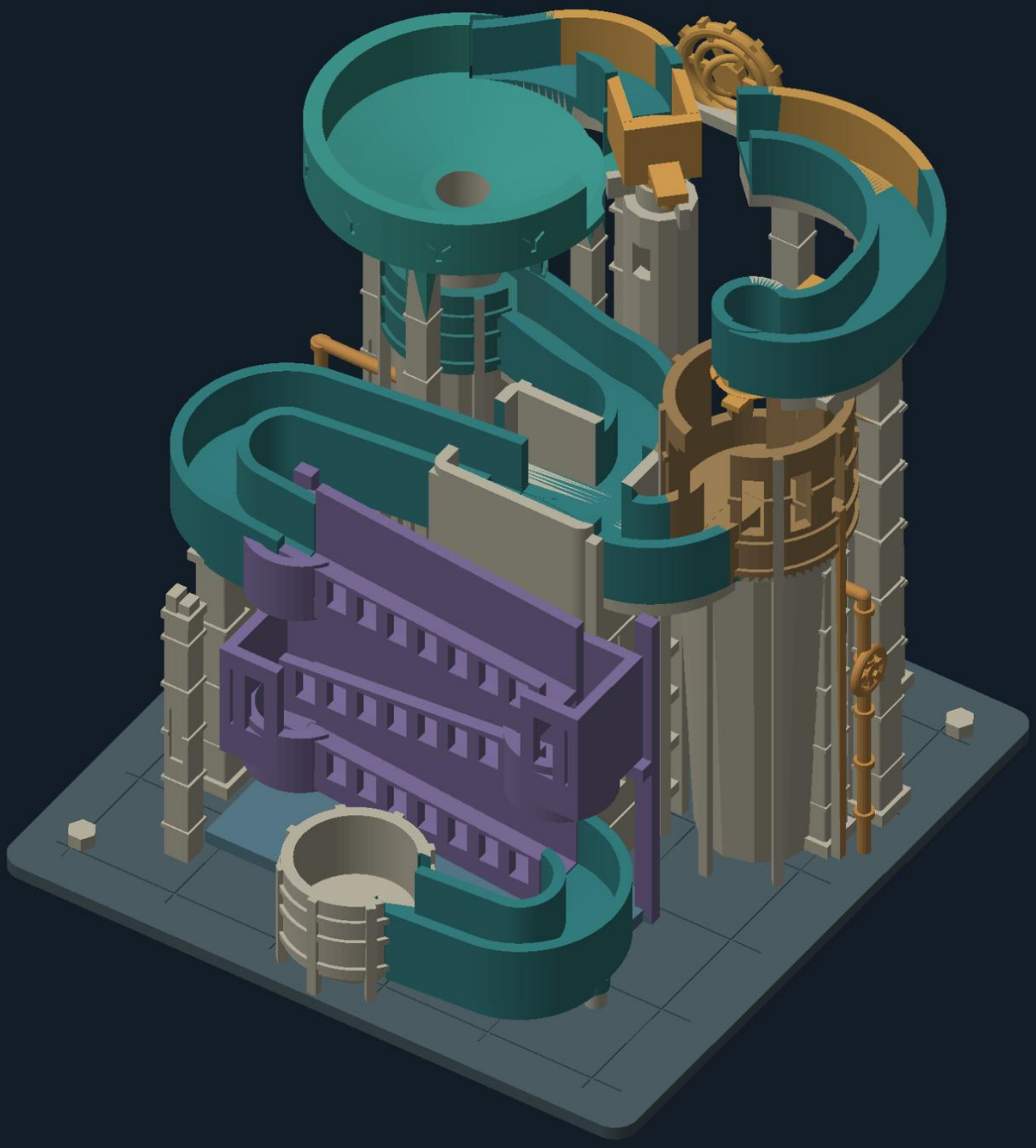
THE LAST OBSERVATORY / PRISM MACHINE

Manual selector · funnel · tipping cup · shared guarded cascade · front collector



Actual separate-part CAD assembly · 144 mm base · nominal 8 mm printed marble

Review geometry. Physical rolling, tipping, reset, supports and print fit remain untested.

A STAR CHOOSES ITS DESCENT

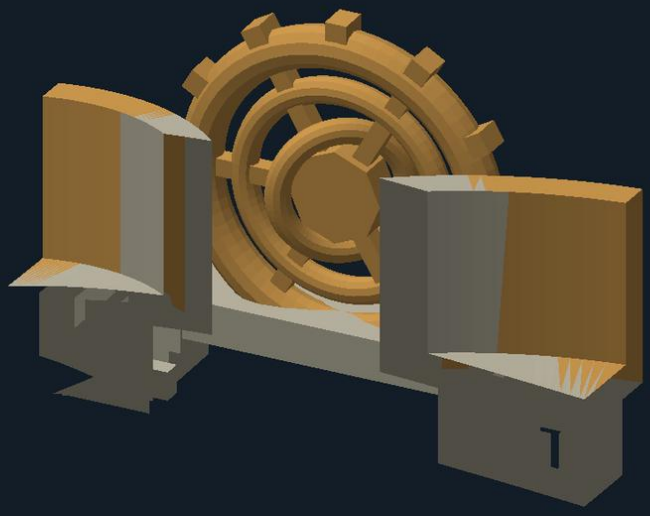
Set the manual selector. One branch spirals through a funnel; the other loads a tipping cup. Both return to the shared hall, descend through three staggered trays and two guarded drops, then arrive at the front collector.

Carved hall niches, broken stonework, celestial runes, prominent conduits and an armillary crest tell the Observatory story. Original CAD for Yonathan to review; nominal 8 mm PRINTED marble and 144 mm square base.

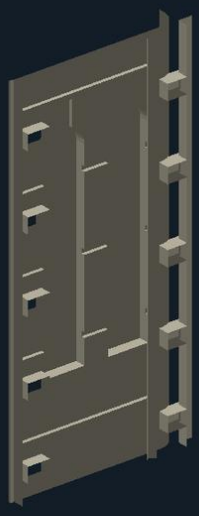
THE OBSERVATORY / CARVED RUINS & INSTRUMENTS

Fixed story details in the actual CAD. Inspection crops omit surrounding context.

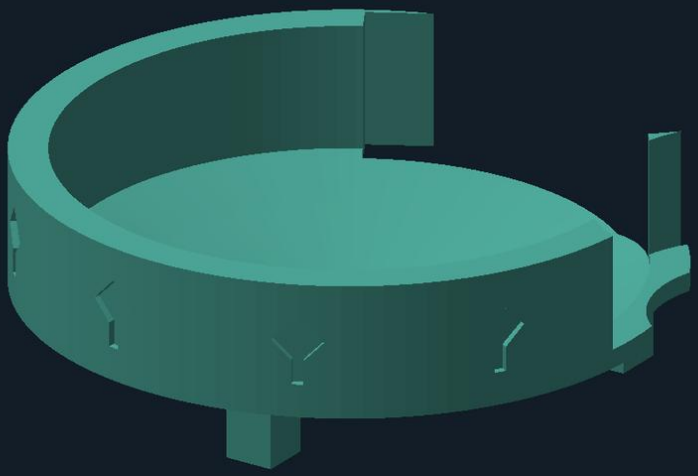
ARMILLARY / MERIDIAN / SURVEY CREST



CARVED HALL / DRESSED STONE NICHES



CELESTIAL RUNES / RETAINING RIM



CONDUIT / FIXED SERVICE GAUGE



The gauge, conduits and instrument are fixed sculpture. Color is an illustrative finishing scheme.

A RUINED INSTRUMENT HOUSE

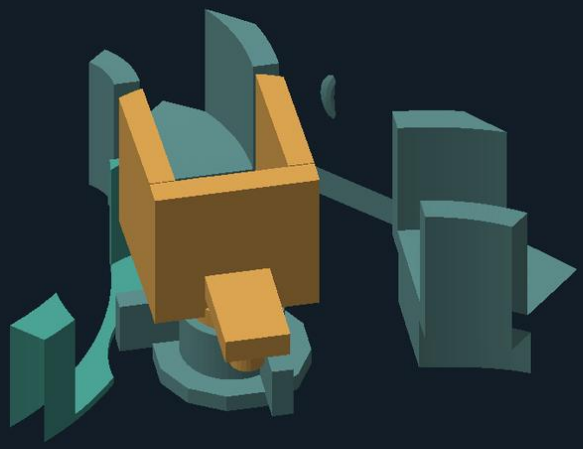
The machinery is held inside dressed-stone towers and a carved machine hall. Recessed celestial runes mark the bowl; lancets and a broken survey pillar suggest the remaining ruins. Raised conduits and a fixed service gauge connect the lower architecture. The richer armillary crest remains above the manual selector.

These details are modeled in CAD. The mechanisms and route retain their approved layout. Fixed sculpture has no powered effects. Mesh, sampled motion and marble-space checks were repeated after the additions.

PRISM MACHINE / MECHANISM INSPECTION

Actual exported meshes. Some fixed context is hidden to expose the mechanisms.

MANUAL SELECTOR / A



MANUAL SELECTOR / B



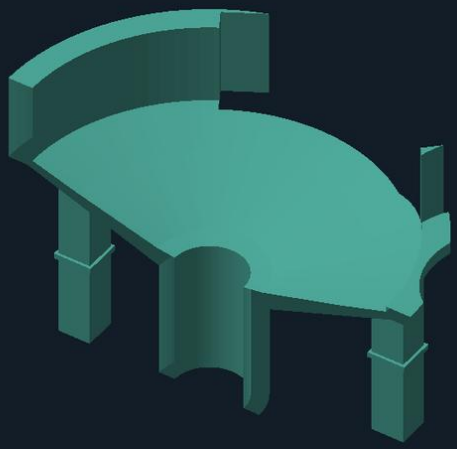
CUP REST POSE / FEED CONTEXT HIDDEN



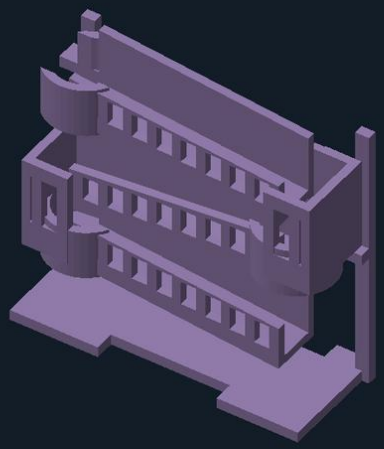
CUP DUMP POSE / FEED CONTEXT HIDDEN



FUNNEL / FRONT HALF HIDDEN



THREE TRAYS / TWO GUARDED DROPS



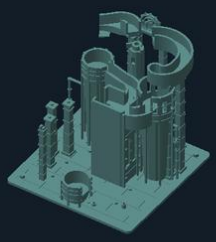
The poses and crop views explain the CAD. They do not show a physical run or certify retention.

Inspection crops hide some fixed context. Shown poses explain the mechanism; rolling, tipping and empty reset remain physically untested.

SEPARATE PRINT PARTS / 14 STL DESIGNS

Each tile has its own scale. Print one of each installed part; retainers ×2. Coupon/alternative as needed.

CHASSIS



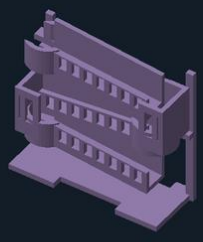
144.0 × 144.0 × 149.8 mm

FUNNEL



61.3 × 59.6 × 131.0 mm

CASCADE



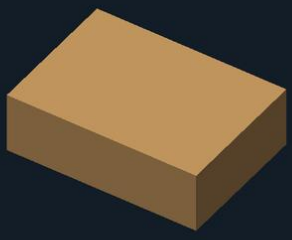
86.4 × 32.4 × 80.0 mm

CUP



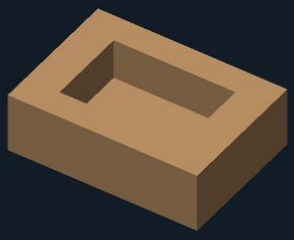
15.8 × 40.1 × 6.0 mm

BALLAST



14.0 × 10.0 × 6.9 mm

BALLAST-LIGHT



14.0 × 10.0 × 6.9 mm

BALANCE-ENDCLIP



6.4 × 3.2 × 3.2 mm

SELECTOR



15.2 × 28.0 × 18.4 mm

AXLE



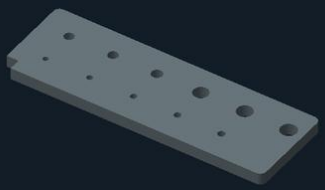
1.8 × 1.8 × 18.8 mm

RETAINER ×2



4.2 × 4.2 × 2.4 mm

COUPON



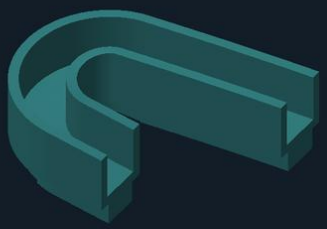
90.0 × 30.0 × 4.0 mm

MARBLE



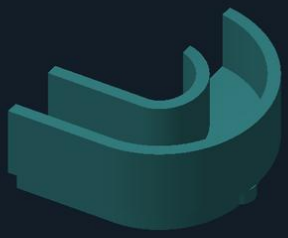
8.0 × 8.0 × 8.0 mm

APPROACH-BRIDGE



64.6 × 41.2 × 31.0 mm

COLLECTOR-RAMP



40.9 × 41.2 × 25.0 mm

Assembly STL is view-only. STL meshes are not a sliced or physically validated print package.

Use the separate part meshes. The assembled STL is for viewing only. Parts have individual scales in this sheet.

ASSEMBLY / DRY-FIT BEFORE PRINT COMMITMENT

1 Coupon and surface finish

Print the fit coupon first. The notched front-left corner identifies its rows: Y=9 gives 1.8/1.95/2.1/2.25/2.4 mm holes; Y=21 gives 3.2/3.4/3.6/5.0/5.2/5.4 mm holes, increasing left to right. Test your actual axle and pegs.

2 Cascade and service rails first

Leave funnel and selector off. Hold cascade 35 mm forward and 68 mm above its seat; slide rearward 35 mm, then lower onto four pegs at X72,Y42,Z3. Hold collector ramp 6 mm high and 20 mm right, slide left, then lower. Lower the approach bridge onto its three pier keys. The marble course retains its approved layout.

3 Funnel and manual selector

Lower the funnel at X38,Y104,Z3, aligned with branch A. Then place the selector stem at X75.5,Y116. Its 4.6 mm stem fits a nominal 5.0 mm bearing. Align 225 degrees for A or 135 degrees for B; position before feeding. For removal: selector, funnel, bridge, ramp, then cascade; reverse the installation motions.

4 Capture the cup axle

Cup bowl faces forward; counterweight fork faces rearward. Axle centre: X115.411,Y104,Z114. Thread a nominal 1.75 x 18.8 mm axle through posts and cup. Fit two retainers outside the posts. Never force a brittle retainer. Verify free motion and positive retention.

5 Fit and tune the counterweight

Slide the regular carrier into the rear fork, with its neck and underside flange straddling the lever. Fit the endclip. Begin with the carrier centre 9.7 mm behind the axle; reviewed range 9.5–10.5 mm. Secure a loose clip/slider with a small removable tape tab outside contact surfaces.

6 Feed one solid printed marble

Use a nominal 8 mm PLA printed marble, initially 100% infill. Weigh and measure it. The ideal solid ball is about 0.33 g. A lighter ball may need the optional light carrier or manual reset. Do not install both weights together. No glass/steel substitution is assumed.

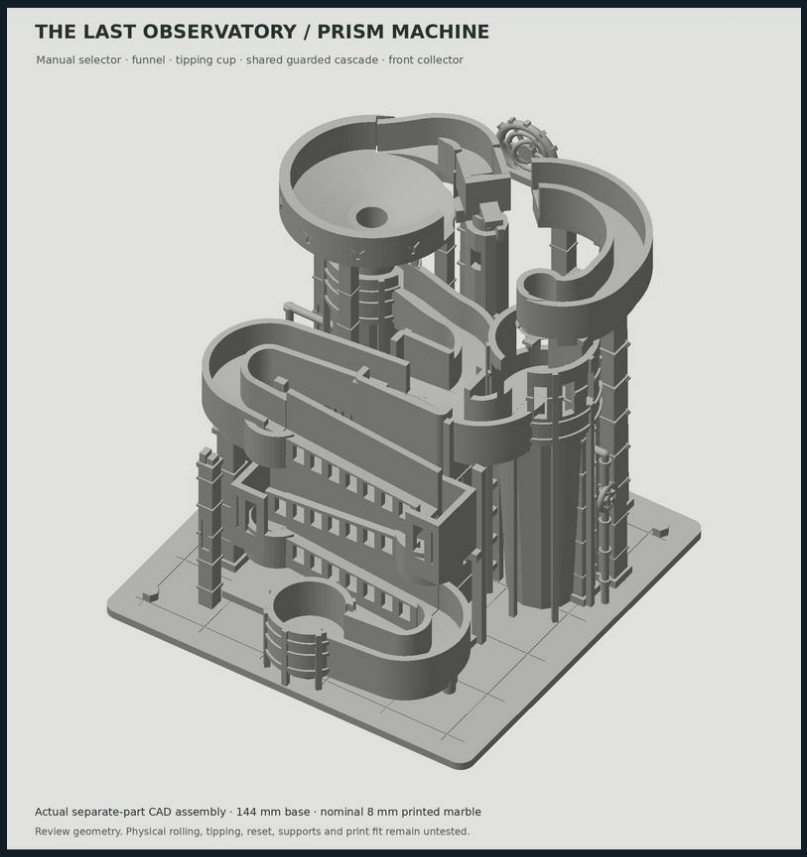
Full instructions, dimensions and reproduction commands are in START-HERE.txt. Dry-fit all seats; tune with measured masses and your actual print tolerances.

MEASURED DIGITAL CHECKS

Printable meshes	14 / 14 topology checks: PASS
Mesh conditions	One closed connected solid each; oriented edges; no degenerate faces
Envelope	144 x 144 x 149.8 mm chassis; 3 mm foundation
Stationary assembly	Funnel/chassis, cascade/chassis and module/module surface crossings: PASS
Sampled motion	60 cup + 180 carrier + 90 selector interior poses: PASS
Assembly / removal	451 reversible path poses; keyed seats and ten fixed-part pairs
Marble envelopes	2,344 static 8 mm sphere probes: PASS
Gravity estimate	Ideal solid-PLA loading/restoring signs: PASS

The marble screen covers both selector interfaces, all six fixed tracks with +/-0.6 mm lateral offsets, bowl/throat, cup loading and release construction, both merge approaches, cascade inlet, two guarded drops, final outlet and collector. Sphere contact heights are solved against exported triangles.

A 0.04 mm contact offset is an analysis construction, not manufacturing clearance. Spiral and falling paths are geometric possibilities, not predicted trajectories. Motion uses noncoplanar surface crossings at 1-degree samples. Five keyed seats were measured from actual meshes; shared-plane contacts within 0.0002 mm are recorded as export precision. Continuous-time forces, printer fit and wear are not certified.



PRINT REVIEW / PHYSICAL TESTS STILL REQUIRED

Slicing and supports

Print at 100% scale. Check thin 0.8 mm cup walls survive slicing. Keep all channels open upward. Elevated chassis rails, tall funnel legs, the cup hub and selector stem need a deliberate support/adhesion strategy. Inspect every layer and removal access. No sliced profile, G-code, print-time estimate or completed print is supplied.

Mechanism validation

Nominal cup stops: -5 degrees rest, +55 degrees dump. The low, broad carrier gives the intended gravity signs in the ideal calculation, including empty return. It ignores axle friction, actual infill/density, impacts and wear. Weigh parts, tune balance and test triggering, release, retention and reset. Automatic operation is not proven.

Test in stages

On a level surface, feed one ball slowly. Check each selector exit, funnel circulation and throat, cup loading/release, both merge approaches, each guarded drop, final feed and collector. Stop after any stall or escape. Record actual observations and revise the CAD/finish as necessary. Static clearance does not prove rolling speed, bounce or retention.

Authorship and programme status

Original CAD generated by OpenAI Codex for Yonathan to review, following his approved three-action concept and printed 8 mm baseline. Earlier projects/revisions are preserved. No student work hours, manual CAD authorship, native Onshape history, eligibility, acceptance or physical success is claimed. Programme AI rules and account-specific submission requirements remain unverified.

Included evidence

Editable OpenSCAD, generated route data, 14 separate STL designs, 8 mm ball, fit coupon, raw chassis export and precision-cleanup report, assembly placement manifest, exact probe CSV/JSON, motion/topology and assembly-path reports, source rendering tools and this PDF. Nothing has been submitted, publicly published or printed.

Guide: github.com/karnxx/merbl/blob/main/guide.md#marble. Mechanism references: Woodgears marble elements/flip-flop and Out Of Marbles funnel/bucket descriptions; no commercial CAD geometry copied.